Computer Vision and Image Understanding 260 (2025) 104459

Contents lists available at ScienceDirect

Computer Vision and Image Understanding

journal homepage: www.elsevier.com/locate/cviu

Robust cross-image adversarial watermark with JPEG resistance for defending against Deepfake models

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ARTICLE INFO

Communicated by Zhun Zhong

Keywords:
Deepfake
Proactive defense
Adversarial attack
JPEG-resistance
Cross-image

ABSTRACT

The widespread convenience of generative models has exacerbated the misuse of attribute-editing-based Deepfake technologies, leading to the proliferation of illegally generated content that severely threatens personal privacy and security. Existing proactive defense strategies mitigate Deepfake attacks by embedding imperceptible adversarial watermarks into the spatial-domain of protected images. However, spatial-domain adversarial watermarks are inherently sensitive to lossy compression operations, which significantly degrades their defense efficacy. To address this limitation, we propose a frequency-domain cross-image adversarial watermark generation scheme to enhance robustness toward JPEG compression. In the proposed method, the adversarial watermark training process is migrated to the frequency domain using a differentiable JPEG module, which explicitly simulates the impact of quantization and compression on perturbation distributions. Furthermore, a fusion module is incorporated to coordinate watermark distributions across images, thereby enhancing the generalization of the defense. Experimental results demonstrate that the generated adversarial watermarks exhibit strong robustness against JPEG compression and effectively disrupt the outputs of Deepfake models. Moreover, the proposed scheme can be directly applied to diverse facial images without retraining, thereby providing reliable protection for real-world image application scenarios.

1. Introduction

Deep learning models have greatly advanced generative image technologies, with one of their most prominent applications being facial manipulation, commonly known as Deepfake (Nguyen et al., 2022; Seow et al., 2022). Deepfake models for face attribute editing and face swapping enable the intentional creation of highly realistic facial images and videos (Tolosana et al., 2020; Juefei-Xu et al., 2022; Yin et al., 2023). However, the misuse of this technology poses significant risks to individuals and society. Fabricated images with exceptional visual fidelity can undermine personal privacy when exploited by malicious users and mislead public perception when widely disseminated, creating potential for social disruption. Addressing the misuse of such technologies remains an urgent challenge.

To address the security threats posed by Deepfake technology, two primary defense strategies have been proposed — passive detection and proactive defense — as depicted in Fig. 1(a). The passive detection approach is typically implemented after forged data have been generated, relying on the training of Deepfake detectors to distinguish counterfeit images via binary classification or feature consistency analysis (Chen et al., 2021; Tariq et al., 2021; Qiao et al., 2023; Zhuang et al., 2022; Zhang et al., 2024). Recent advancements in this area include methods for detecting post-processed image forgeries through signal noise separation (Chen et al., 2022) and identifying compressed Deepfake videos by analyzing facial muscle motions (Liao et al., 2023), enhancing the robustness of passive detection against various manipulations. However, this method is inherently limited by a time lag, which impedes the prompt prevention of the malicious dissemination of forged images and, consequently, fails to mitigate the resultant harm. In contrast, proactive defense mechanisms (Ruiz et al., 2020; Huang et al., 2021, 2022; Lin et al., 2022; Tang et al., 2023; Xu et al., 2024; Qu et al., 2024; Qiao et al., 2024) involve embedding imperceptible adversarial watermarks into protected images. This strategy compels GAN models to produce perceptibly distorted images when attempting to forge them, thereby effectively preventing the malicious use of forged images and providing a more proactive form of protection.

Existing research has progressively demonstrated the feasibility of adversarial attacks within proactive defense mechanisms. Ruiz et al.

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https://doi.org/10.1016/j.cviu.2025.104459
Received 28 March 2025; Received in revised form 23 July 2025; Accepted 4 August 2025
Available online 11 August 2025

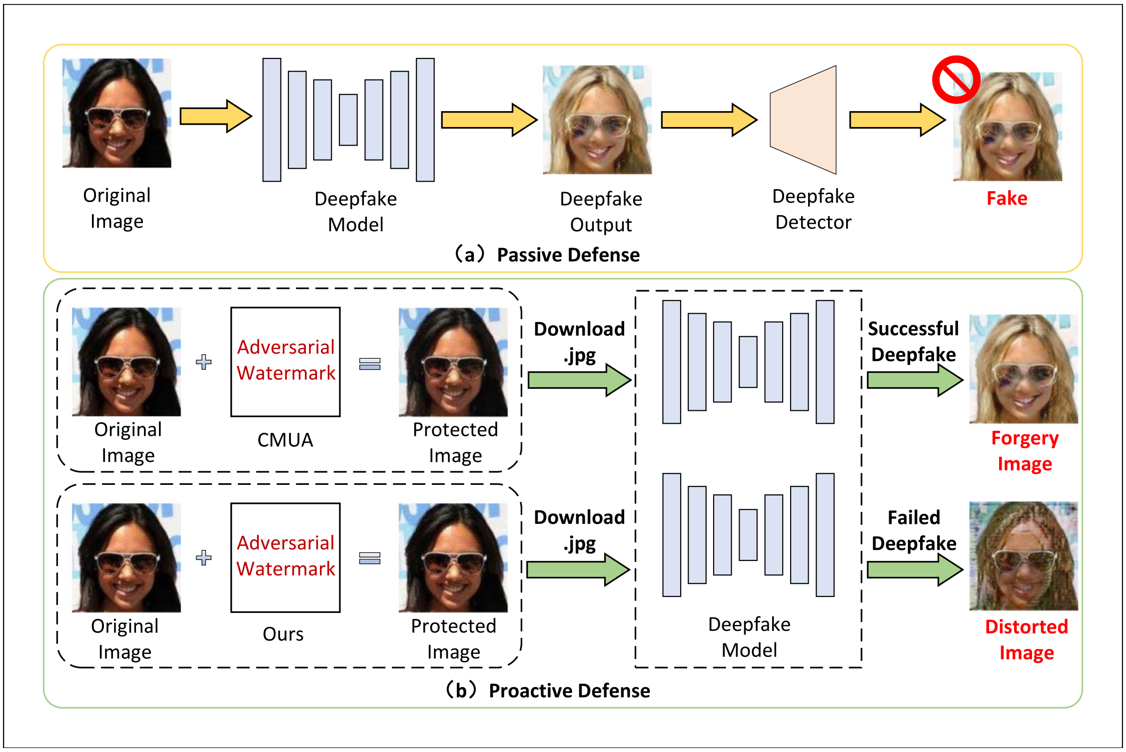

Fig. 1. (a) Passive detection mechanisms perform authenticity verification only after forgery completion, thereby exhibiting inherent limitations in post-hoc forensics. (b) Proactive defense mechanisms counteract forgery processes by embedding adversarial watermarks at the propagation source. However, conventional spatial-domain watermarks fail under JPEG compression due to high-frequency sensitivity. In contrast, the proposed method preserves defense efficacy under JPEG compression, effectively disrupting Deepfake content generation.

(2020) were the pioneers in experimentally validating this theoretical framework, with a focus on assessing the effectiveness of defense strategies. Building upon this foundational work, Huang et al. (2022) introduced the Cross-Model Universal Adversarial (CMUA) watermark method, which enhances the transferability of watermarks through novel training strategies, thereby enabling cross-model protection against a range of forgery models. To further extend scalability, Qiao et al. (2024) developed the Scalable Universal Adversarial (SUA) watermark method, effectively addressing emerging forgery models without the need for model retraining. While these existing approaches exhibit strong defensive performance in controlled testing scenarios, they demonstrate insufficient robustness against image compression, particularly in the JPEG format. As illustrated in Fig. 1(b), this study shows that current methods, exemplified by CMUA, experience significant performance degradation when subjected to JPEG compression.

As a widely adopted standard for image compression, JPEG operates through two essential processing stages: (1) applying the Discrete Cosine Transform (DCT) to input images, and (2) performing quantization on the resulting DCT coefficients (Shin et al., 2017; Reich et al., 2024). Pixel-domain adversarial watermarking methods (e.g., CMUA) typically rely on subtle pixel-level perturbations, particularly susceptible to amplitude attenuation caused by rounding operations during the quantization phase. This issue is especially pronounced in high-frequency DCT coefficient regions. Such attenuation leads to feature distortion of pixel-level embedded adversarial watermarks in the reconstructed compressed images, ultimately undermining the effectiveness of adversarial watermarks in JPEG compression scenarios.

To address the potential impact of JPEG compression on adversarial watermarks, this paper presents an effective solution. We construct adversarial watermarks in the DCT frequency domain and incorporate differentiable JPEG compression to maintain the differentiability of the JPEG process during end-to-end training. This design allows for gradient backpropagation, which adaptively guides the generation of robust perturbation distributions within the DCT domain. Additionally, we propose a coordination mechanism to resolve conflicts between gradient propagation and watermark fusion during multi-image adversarial watermark generation, thereby enhancing the cross-image transferability of adversarial watermarks. Experimental results demonstrate that our method significantly improves the robustness of adversarial watermarks against JPEG compression, while effectively protecting diverse facial image datasets. Our contributions can be summarized as the following:

- We propose a JPEG-resistant proactive watermarking defense against Deepfake models by constructing adversarial watermarks in the DCT frequency domain, ensuring their robustness and effectiveness even after JPEG compression.
- We introduce a coordination mechanism to resolve gradient and fusion conflicts in cross-image watermark generation, significantly enhancing the watermark's transferability across different images.
- Extensive experiments demonstrate the efficiency and practicality of the proposed method in mitigating JPEG compression effects and ensuring robust cross-image protection, providing a strong defense against Deepfake models.

2. Related work

2.1. Adversarial attack on Deepfake models

Current Deepfake techniques primarily implement facial manipulation through two technical paradigms: face swapping and attribute modification. This study focuses on the latter, characterized by leveraging GAN-based architectures for targeted semantic editing of facial attributes. These methods utilize conditional image-to-image translation frameworks (e.g., pix2pixHD (Wang et al., 2018)) or cross-domain mapping mechanisms (e.g., CycleGAN (Zhu et al., 2017)) to precisely manipulate localized facial attributes (e.g., age, expression), producing forged samples with high visual fidelity that closely approximate the perceptual quality of authentic images. This fidelity is achieved by preserving facial texture, lighting consistency, and spatial coherence across manipulated regions.

In the realm of adversarial attack and defense research, Wang et al. (2020a) and Yeh et al. (2020) pioneered the establishment of adversarial attack evaluation benchmarks for image translation tasks. Notably, Yeh et al. (2020) first demonstrated the disruptive effects of adversarial examples on attribute modification pipelines in Deepfake generation scenarios. Building on these foundational works, Ruiz et al. (2020) systematically analyzed mainstream facial editing networks, revealing that they predominantly adopt conditional GAN architectures—an observation that has spurred the systematic deployment of adversarial attacks for proactive Deepfake defense. With the rapid advancement of diffusion models (Ho et al., 2020; Croitoru et al., 2023), adversarial attack research targeting these high-quality generative architectures has increasingly garnered significant attention (Liang et al., 2023; Shan et al., 2023; Van Le et al., 2023). For facial attribute modification attacks in Deepfake, this study focuses on defending against three representative approaches: StarGAN (Choi et al., 2018), which enables cross-attribute editing through multi-domain label encoding in a single generator, preserving visual fidelity even with complex attribute modifications, AttGAN (He et al., 2019), which achieves precise localized attribute manipulation by constraining the latent space via an attribute decoupling-reconstruction mechanism, and HiSD (Li et al., 2021), which overcomes uncontrolled manipulations in image-to-image translation by organizing labels into a hierarchical structure of independent tags, exclusive attributes, and disentangled styles for controllable and accurate translations. These methods represent the technical frontiers of facial attribute manipulation, encompassing single-attribute targeted modification, multi-domain translation via shared latent spaces, and disentangled multi-attribute collaborative editing, providing a robust theoretical and experimental benchmark for evaluating the generalization of defense methods.

2.2. Proactive defense methods

Existing researches have demonstrated the feasibility of adversarial attacks against GAN-based models by embedding adversarial watermarks in protected images to achieve proactive defense against Deepfake. Notably, Ruiz et al. (2020) and Yeh et al. (2020) first confirmed that adversarial attacks can effectively disrupt the outputs of conditional image translation face editing networks. Huang et al. (2021) deployed defenses in gray-box scenarios by training surrogate models. To enhance the generalizability of adversarial watermarks, Huang et al. (2022) proposed CMUA, which extends the defense coverage across multiple models. Building upon CMUA, the ACS (Lin et al., 2022) framework optimizes multiple GAN models jointly to eliminate sensitivity to optimization order. Tang et al. (2023) proposed FOUND, a novel two-stage Deepfake disruption method that targets feature extractors across multiple models and applies gradient-ensemble strategies, achieving superior disruption efficiency and universality. Additionally, Qiao et al. (2024) introduced SUA, which uses a three-stage training strategy "inheritance-defense-constraint" to achieve scalable, universal watermark generation. Although these aforementioned proactive defense methods and even later ones like SUA effectively defend against Deepfake editing, their performance significantly deteriorates when protected facial images are subjected to JPEG lossy compression. Recognizing this critical vulnerability exposed by earlier defenses, Qu et al. (2024). proposed DF-RAP, which employs ComGAN to model lossy compression processes and generates adversarial perturbations accordingly, further exploring the robustness of proactive defense mechanisms under compressed transmission channels. Given the potential threat posed by JPEG compression to proactive defenses, our study aims to enhance robustness against JPEG compression by investigating the fundamental principles of the JPEG compression pipeline, thereby improving defense performance in real-world applications.

2.3. JPEG-resistance embedding methods

The potential interference effects of lossy compression algorithms (e.g., JPEG) on adversarial examples have attracted significant research interest. Existing approaches primarily focus on enhancing the robustness of adversarial examples against compression perturbations. For instance, some studies (Wang et al., 2020b; Liu et al., 2023) employ differentiable approximation operations to simulate the non-differentiable JPEG compression process, thereby ensuring that generated adversarial perturbations maintain their attack effectiveness following JPEG compression. Recognizing that JPEG compression achieves data reduction by truncating high-frequency coefficients in the Discrete Cosine Transform (DCT) domain, alternative methods (Sharma et al., 2019; Zhu et al., 2023) restrict adversarial perturbations to low-frequency DCT components to mitigate compression-induced degradation. Similarly to our work, Sielmann and Stelldinger (2023) proposed directly constructing adversarial perturbations in the DCT coefficient domain—a strategy that has been shown to substantially improve the compression robustness of adversarial examples. Aladwan et al. (2024) investigated steganography-enhanced adversarial attacks by embedding perturbations in the frequency domain using DCT, wavelet transforms, and JPEG compression, thereby demonstrating robustness against conventional adversarial defense mechanisms. While robustness against digital compression has been widely studied, ensuring reliable information embedding under physical attacks such as screen-shooting is equally critical. Recent works have explored screen-shooting watermarking leveraging frequency-domain techniques (Fu et al., 2024) or simulating distortions including JPEG compression to enhance robustness (Li et al., 2024).

However, these methods primarily address the design of robust adversarial examples for classification models or general watermarking. In contrast, for our image generation task, the protected image must effectively resist attacks from Deepfake models. Therefore, in this paper we propose a frequency-domain adversarial watermark generation scheme to effectively disrupt the outputs of Deepfake models and enhance robustness toward JPEG compression.

3. Methodology

3.1. Problem formulation

This section provides a detailed description of the adversarial watermark-based proactive defense mechanism. The core concept of proactive defense is to introduce structured distortions that actively disrupt the output of the target Deepfake models. Specifically, the method targets facial attribute editing models based on Conditional Generative Adversarial Networks (Conditional GANs), with the input–output relationship formally defined as: $x' = G(x, c)$, where G represents the forgery model, c denotes the target attribute label, x and x' represent the original face image and its forged counterpart, respectively. By embedding an adversarial watermark w into the protected image, we generate an adversarial example $x_{adv} = x + w$. When x_{adv} input into G, the output $G\left(x_{adv}, c\right)$ significantly deviates from the original forgery result $G(x, c)$, effectively rendering the forged content practically useless. This process can be modeled as the following constrained optimization problem:

$$\max_{w} \ell\left(G(x, c), G\left(x_{\text{adv}}, c\right)\right), \quad \text{s.t.} \quad \|w\| \le \epsilon \tag{1}$$

where ℓ represents the distance loss (typically Mean Squared Error, MSE), and ϵ is the constraint on the magnitude of the adversarial watermark. Based on this optimization objective, adversarial watermarks are generated using attack algorithms such as Projected Gradient Descent (PGD). The iterative update process is expressed as:

$$\begin{aligned} x_0 &= x + w_0 \\ x_{i+1} &= \text{clip}_{x,\epsilon}\left\{x_i + \alpha \cdot \text{sign}\left[\nabla_x \ell\left(G(x, c), G(x_i, c)\right)\right]\right\} \end{aligned} \tag{2}$$

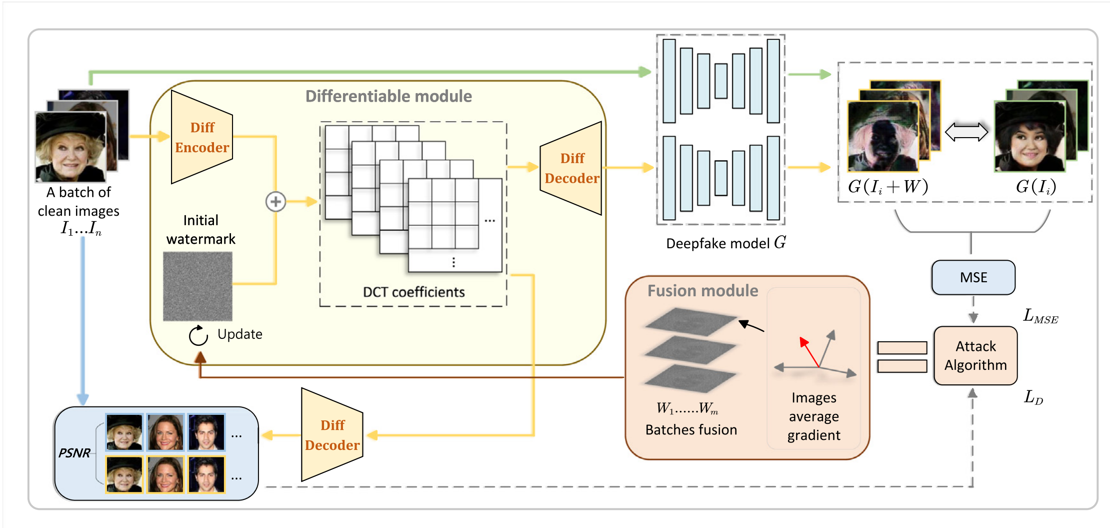

Fig. 2. The overall pipeline of our robust cross-image adversarial watermark defending against Deepfake models.

where w_0 is initialized as a random noise vector (according to PGD), i represents the iteration index, and the clip function ensures that x_i remains within the prescribed range $[x-\epsilon, x+\epsilon]$, α denote the gradient update step size. For clarity in subsequent sections, the output of forgery model will be denoted by $G(x)$.

3.2. The proposed JPEG-resistance adversarial watermarking

To enhance the defense performance of our method under JPEG compression, we embed adversarial watermarks as frequency-domain coefficients within the image transformation domain, thereby improving their robustness to JPEG compression. Additionally, we integrate a differentiable JPEG module into the end-to-end training process, enabling the gradient to flow through the compression operations. This differentiable module overcomes the gradient computation challenge introduced by JPEG compression. By utilizing a specially designed loss function, we allow gradients to propagate through the JPEG compression, thereby enabling the adversarial watermarks to adapt to the compression process. Furthermore, the training process incorporates gradient averaging, which focuses on shared facial attributes across images within the same batch. An inter-batch fusion strategy is also employed to link adversarial watermarks across different batches, thereby enhancing both their defensive capability and cross-image transferability. As illustrated in Fig. 2, we iteratively apply the following procedure to each training batch: a small batch of clean facial images $I_1 \cdots I_n$ is first transformed into their corresponding DCT frequency components using a differentiable encoder. A random noise vector W is then added to these DCT components, serving as the initial adversarial watermark. These frequency components are subsequently transformed back into the spatial domain using a differentiable decoder, yielding the adversarial watermarked images, which, together with the clean images, is input into the Deepfake model to generate the preliminary disrupted images $G(I_i+W)$ and $G(I_i)$.

The perceptual loss computes the difference between I_i+W and I_i, while the MSE loss calculates the discrepancy between $G(I_i+W)$ and $G(I_i)$, thereby yielding the total loss value. The average gradient across different images is then used to guide the PGD algorithm in updating the adversarial watermark within the current batch. Upon completing the update of the adversarial watermark for each batch, a batch fusion strategy is employed to effectively integrate the adversarial watermarks generated by different batches. Upon the completion of the training process across all batches, the resulting adversarial watermark is capable of providing robust protection for a large set of facial images. The subsequent sections systematically detail the implementation of this transformation process using differentiable modules, in conjunction with multi-image fusion strategies, to generate the proposed JPEG-resistance adversarial watermark.

3.2.1. The differentiable JPEG module

In contrast to traditional spatial-domain adversarial watermarking methods, we incorporate a differentiable JPEG process into our framework, utilizing frequency-domain transformation. This approach enables adversarial watermark optimization in the frequency domain, guided by attack algorithms and a specific loss function, while directly embedding the watermark into the image's frequency-domain representation. Due to the differentiable nature of this process, the quantization errors introduced by JPEG compression are accounted for in each iteration, thereby directing the optimization of the adversarial watermark in the form of DCT coefficients. This feature allows the attack algorithm to generate more robust adversarial watermarks.

In standard JPEG compression, the quantization of DCT coefficients typically relies on a non-differentiable rounding function, whose gradient is nearly zero across most regions. This renders it unsuitable for gradient-based optimization methods commonly used in adversarial perturbation generation. To address this limitation, prior work has introduced differentiable JPEG simulation modules (e.g., DiffJPEG (Shin et al., 2017)), which replace the non-differentiable rounding operation with smooth approximations to enable gradient backpropagation through the compression process. In our method, we adopt the approximation function: $x_{approx} = round(x) + (x - round(x))^3$, and integrate the differentiable JPEG module into our training framework, denoting the encoder and decoder as $E_n(\cdot)$ and $D_e(\cdot)$, respectively. The encoder $E_n(\cdot)$ transforms input images into intermediate DCT representations, while the decoder $D_e(\cdot)$ performs the inverse transformation back to the spatial domain. Owing to the differentiable nature of this module, quantization-induced compression errors are incorporated into the gradient computation, allowing the adversarial watermark to be optimized in the frequency domain with improved robustness against compression.

We convert images into DCT frequency-domain representations through a series of operations, including color space conversion, chrominance subsampling, DCT transformation, and differentiable quantization. Specifically, we integrate a JPEG differentiable encoder $E_n(\cdot)$ to transform a batch of clean images I into intermediate DCT coefficients. These DCT coefficients subsequently replace the clean images for further processing. The color space conversion transforms RGB-formatted clean images into the $[Y, cb, cr]$ components, which are then processed separately. This transformation theoretically allows for the generation of three intermediate DCT coefficients through $E_n(I)$. However, in this study, we preserve the cb and cr components unchanged, and apply DCT coefficients only to the Y-component (luminance). Therefore, we focus exclusively on the Y component, denoting the DCT coefficients as y, with the relationship $y = E_n(I)$.

Furthermore, in the frequency-domain representation, we add the DCT-coefficient adversarial watermark, denoted as W, to the intermediate DCT coefficients of the images. This process results in watermarked intermediate DCT coefficients. Finally, we use a JPEG differentiable decoder $D_e(\cdot)$ to perform the inverse transformation, which denoted $D_e(y+W)$, converting the intermediate DCT coefficients back into spatial-domain images. For proactive defense, the adversarial watermark embedded in facial images must simultaneously satisfy two critical properties: the ability to defend against GAN forgery models and the imperceptibility of the adversarial watermark. Therefore, in our method, we incorporate perceptual loss into the adversarial watermark optimization objective to ensure that the inverse-transformed image I^{inv} adheres to the imperceptibility constraint: $I^{inv} = D_e(y+W) \approx I$. During each iteration, the total loss, including perceptual loss $L_D = PSNR\left(I^{inv}, I\right)$, is calculated for backpropagation, thereby forming the complete loss function:

$$L = L_{MSE}\left(G\left(I^{inv}\right), G(I)\right) + \lambda \cdot 1/L_D \tag{3}$$

where L_{MSE} represents the MSE loss between adversarial and original outputs of Deepfake model G, and L_D is computed based on the peak signal-to-noise ratio (PSNR), a widely used metric for visual perception. The λ values are hyperparameters used to balance the terms.

Finally, we employ the PGD attack algorithm to generate adversarial watermarks. The formulation in Eq. (2) can be directly applied to the frequency-domain scenario as follows:

$$\begin{aligned} y^0 &= y + W_{init} \\ y^{t+1} &= clip_{y,\epsilon}\left\{y^t + \alpha \cdot sign\left(\nabla_y L\right)\right\} \end{aligned} \tag{4}$$

where y^t denotes the intermediate DCT coefficients of the facial images at the tth iteration, L represents the loss function as defined in Eq. (3), and W_{init} corresponds to the random initial adversarial watermark introduced before the iterative optimization process.

However, the adversarial watermarks generated through the aforementioned procedure exhibit specificity to individual facial images. To address this limitation, we dedicate the following subsection to methodologies aimed at enhancing their generalizability across multiple images.

3.2.2. Cross-image fusion module

The adversarial watermarks generated from different images exhibit significant divergence due to conflicts arising from facial feature heterogeneity, which may reduce the defensive capability of the final watermarks. To address this issue, we employ two strategies. Specifically, when training a batch of facial images, the average gradient across different images is utilized:

$$\overline{grad} = \frac{1}{n}\sum sign\left(\nabla_{I_i} L\left(G\left(I_i^{wm}\right), G\left(I_i\right), I_i^{wm}, I_i\right)\right) \tag{5}$$

where n denotes the batch size, and I_i^{wm} represents the ith adversarial image in the batch. This approach emphasizes common attributes across different facial images. We then update the batch-specific adversarial watermark w via the PGD attack algorithm as described in Eq. (4) using $\overline{grad}$.

Based on the obtained w, we apply a batch fusion strategy to link the watermarks $w_1 \dots w_m$ acquired from each training batch of facial images. The initial watermark W^0 is derived from the adversarial watermark trained on the first batch:

$$\begin{aligned} W^0 &= w^0 \\ W^{m+1} &= \beta \cdot W^m + (1-\beta) \cdot w^m \end{aligned} \tag{6}$$

where m indicates the number of training batches, W^m denotes the fused adversarial watermark from the first m batches, and β is a hyperparameter that balances the contributions of previous and subsequent watermarks. This strategy helps preserve the contributions of adversarial watermarks across batches, enabling our method to achieve stronger cross-image transferability while protecting diverse facial images.

To provide a clearer explanation of the overall framework process, the pseudocode is presented in Algorithm 1

Algorithm 1 JPEG-resistance adversarial watermark generation

Input: X(dataset contains m batches of training facial images with batch size n), G(the target Deepfake models), E_n, D_e(the differentiable encoder and decoder), T(iteration number of adversarial attack), A(the adversarial attack algorithm which iteratively updates the image-specific watermark)
Output: JPEG-resistance adversarial watermark W

```
1: Random Init W_init
2: for k ∈ [1, m] do
3:     // Discrete Cosine Transform
4:     y_k ⇐ E_n(x_k)
5:     y_k^0 ⇐ y_k + W_init
6:     w_k1 ∼ w_kn ⇐ A(y_k^0, D_e(y_k^0), G, T)
7:     w_k ⇐Compute average gradient with w_k1 ∼ w_kn
8:     W_{k+1} ⇐Batches fusion with w_1 ∼ w_k
9: end for
10: W = W_{m+1}
```

4. Experimental results

4.1. Experimental setting

(1) Dataset: In this study, we conducted experiments on two publicly available facial datasets: CelebFaces Attributes dataset (CelebA) and Labeled Faces in the Wild dataset (LFW). The CelebA test set comprises 19,962 facial images annotated with 40 attributes, each provided with corresponding binary labels. The LFW dataset contains 13,000 labeled facial images from 5,749 unique identities, with attribute annotations consistent with those in CelebA. Specifically, we utilized the first 128 facial images from each dataset to generate adversarial watermarks, and subsequently evaluated the performance of the proposed method on the remaining test images. All facial images were uniformly resized to 256 × 256 during preprocessing to ensure consistency across datasets.

(2) Target model: We adopt three GAN-based facial editing models as target frameworks: StarGAN, AttGAN, and HiSD. All models utilize pretrained configurations to ensure optimal forgery performance. StarGAN is trained on five commonly used attributes from the CelebA dataset, including black hair, blond hair, brown hair, gender, and age. In contrast, AttGAN exhibits superior capability in flexible and fine-grained attribute manipulation, having been trained on up to 14 facial attributes from the CelebA dataset. We also implemented the same configuration for HiSD as used with the comparative methods, CMUA (Huang et al., 2022) and SUA (Qiao et al., 2024), which is trained on CelebA and can add target attributes to the target face.

(3) Implementation details: During the training phase, the selected 128 facial images were divided into 16 batches, each containing 8 images, followed by iterative optimization over a single epoch. For performance evaluation, 1000 distinct facial images were randomly sampled from non-overlapping test sets. Specifically, performance assessment of StarGAN's five-attribute configuration involved generating 5000 manipulated facial images (1000 per attribute), while AttGAN's extended 14-attribute framework required evaluating 14,000 corresponding synthetic samples. The average metric values across all test

cases were calculated to provide a comprehensive assessment of adversarial watermark performance. For HiSD, the evaluation was conducted on 1000 forged outputs generated from 1000 randomly sampled facial images. The average metric values across all test cases were calculated to provide a comprehensive assessment of adversarial watermark performance. We adopt PGD as our attack algorithm, and set the hyperparameters β to 0.50, λ to 0.05, the step size α to 0.03. For the three target models — StarGAN, AttGAN, and HiSD — we respectively set ϵ to 0.20, 0.18, and 0.30, with the number of attack iterations T set to 15, 15, and 20. The code of our work can be found in http://github.com/fishlin20/JPEG-Watermark.

4.2. Evaluation metrics

To comprehensively evaluate the defensive performance of the proposed method, we adopt several widely recognized metrics to quantitatively assess two critical aspects: defense success rate and visual quality degradation of the protected images. Conventional methods typically quantify defense effectiveness by calculating the L_2 distance between distorted outputs and original images, with a threshold of 0.05 commonly used to indicate successful defense. However, this global computation inherently includes non-target regions, potentially diluting the measured distortion ratio within manipulated areas. Consequently, visually prominent artifacts in forged regions may be misclassified as failed defenses due to insufficient overall metric deviation. To overcome this limitation, Huang et al. (2022) introduced an improved evaluation metric L_2^{mask} that applies artifact-aware weighting to selectively emphasize manipulated regions, enabling more accurate defense success rate SR_{mask} quantification. Our experimental framework strictly adheres to this refined evaluation protocol to ensure precise performance assessment.

Furthermore, as adversarial watermarks achieve proactive Deepfake prevention by introducing visual distortion into manipulated outputs, we evaluate the induced visual quality degradation using three standard perceptual quality metrics: PSNR, Structural Similarity Index (SSIM), and MSE.

4.3. Performance evaluation

This section presents a systematic experimental evaluation of the proposed method. First, we assess its performance using both qualitative and quantitative analyses, followed by comparative studies with several baseline methods. Specifically, the selected baselines include: the pioneering proactive Deepfake defense framework (Ruiz et al., 2020), which first introduced proactive countermeasures against forgery attacks; the first universal adversarial watermarking approach (CMUA) (Huang et al., 2022), where we explicitly compare its performance under cross-model universal conditions with its single-model defense configuration, is respectively denoted as CMUA and CMUA(S); and the scalable defense model training scheme (SUA) (Qiao et al., 2024), which employs extensible adversarial training mechanisms; and the DF-RAP (Qu et al., 2024), which investigates the robustness of proactive defenses under lossy compression by integrating ComGAN to generate adversarial perturbation. All comparative experiments strictly adhere to the original configurations specified in the respective publications. Finally, we empirically validate the cross-image transferability of the proposed method.

4.3.1. Defending performance on Deepfake models

The proposed method is deployed on three Deepfake models, StarGAN, AttGAN and HiSD, and evaluated for proactive defense capabilities on two benchmark datasets, CelebA and LFW. Experimental results are summarized in Table 1. Quantitative analysis reveals that our method achieves excellent defensive performance against both forgery models, with particularly outstanding efficacy in countering StarGAN. Specifically, our method attains an SR_{mask} of 100.00% against StarGAN on both CelebA and LFW, with L_2^{mask} metrics consistently exceeding 0.431, thereby indicating that StarGAN fails to correctly modify the target attributes of images protected by our adversarial watermarks. Furthermore, visual results (see Fig. 3) demonstrate that the adversarial watermarks generated by our method induce severe distortions in the output images, confirming the robust defense efficacy of the proposed approach in suppressing StarGAN's attribute-editing attacks on facial images.

Table 1
The quantitative results of the proposed method.

Dataset	Model	PSNR↓	SSIM↓	MSE↑	L_2^{mask} ↑	SR_{mask} (%) ↑
CelebA	StarGAN	9.72	0.39	0.447	0.489	100.00
	AttGAN	15.10	0.59	0.135	0.232	99.07
	HiSD	25.29	0.91	0.009	0.092	74.30
LFW	StarGAN	10.81	0.43	0.346	0.431	100.00
	AttGAN	13.37	0.40	0.198	0.260	100.00
	HiSD	23.31	0.87	0.009	0.069	53.50

Furthermore, in defense scenarios against AttGAN, although the L_2^{mask} metrics on CelebA and LFW decrease by 52.6% and 39.7%, respectively, compared to StarGAN, these metrics still exceed the empirical threshold (0.05) by factors of 4.64 and 5.20, respectively. This observation suggests that AttGAN inherently exhibits stronger resistance to adversarial watermarks than StarGAN. Nevertheless, our method achieves SR_{mask} values exceeding 99.07% on both datasets, demonstrating robust defensive efficacy even against AttGAN. Compared to StarGAN and AttGAN, our method exhibit slightly lower performance in terms of L_2^{mask} and SR_{mask} when defending against HiSD. However, L_2^{mask} value remains above the threshold, and the defense success rate SR_{mask} is still relatively high, demonstrating the effectiveness of our approach against Deepfake models.

Additionally, as illustrated in Fig. 3, the adversarial watermarks effectively inhibit manipulation attempt by all tested Deepfake models. The PSNR and SSIM metrics of the forged outputs from protected images against StarGAN and AttGAN are reduced to below 15.10 and 0.59, respectively (with lower values indicating more severe distortion), and are accompanied by elevated MSE values. Visualization results indicate that our method produces less pronounced disruption on HiSD in comparison to StarGAN and AttGAN. The performance gap can be attributed to HiSD's architectural design. As a disentangled translation model, HiSD modifies images through abstract style features rather than predefined attributes, making attribute-specific frequency-domain watermarks less effective. Its robust encoder–decoder structure and style modulation further mitigate localized perturbations, as reflected by higher PSNR and SSIM scores. Nonetheless, our method still induces notable L_2^{mask} increases and performance drops, indicating partial effectiveness even against such resilient models.

These results collectively confirm that the generated adversarial watermarks induce irreversible visual quality degradation in the outputs of those Deepfake models, significantly undermining the perceptual integrity of the forged images and rendering them practically unusable. In summary, the experimental results presented in Table 1 validate the universal defense capability of our method against diverse Deepfake models, further substantiating the effectiveness of proactive strategies in countering GAN-based attribute manipulation.

4.3.2. Performance under JPEG compression

This section aims to evaluate the defense performance of the proposed method in comparison with existing baseline approaches under varying JPEG compression quality factors (Q-values). Specifically, we employ the quantitative metrics L_2^{mask} and SR_{mask} to assess performance differences.

The comparison results are presented in Fig. 4. Overall, it can be observed that as the Q-value decreases (i.e., the strength of JPEG compression increases), the defense effectiveness of all methods deteriorates to varying degrees. This indicates that the quantization operation

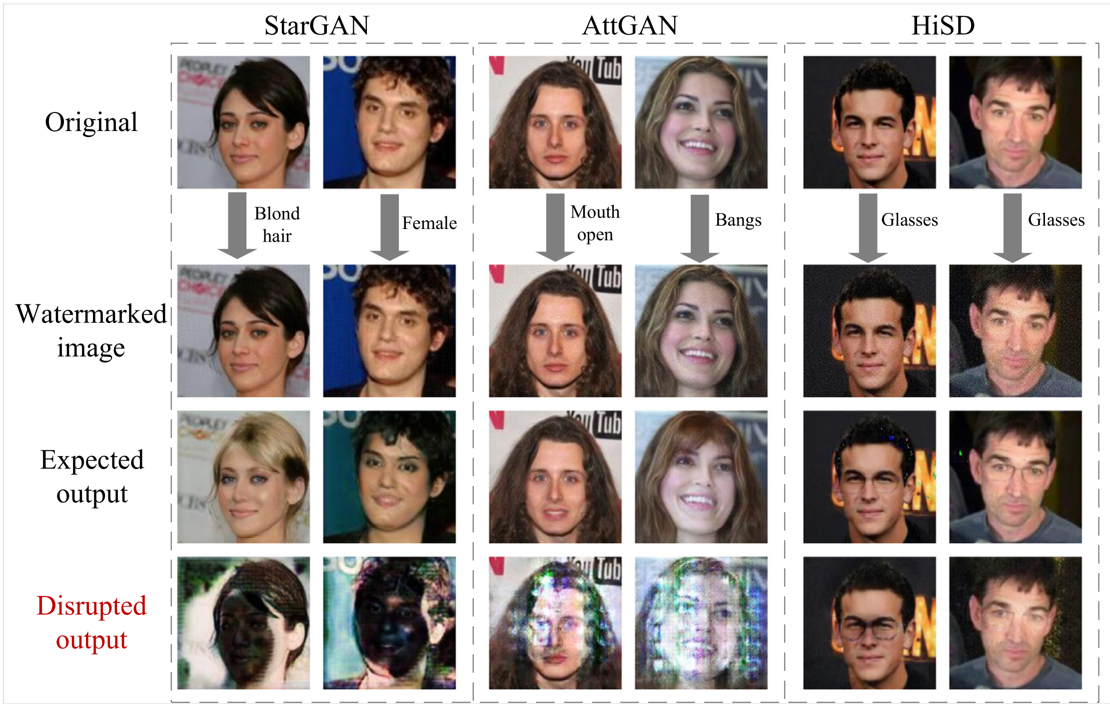

Fig. 3. The image examples under defense performance experiments from different target models applied to the CelebA and LFW datasets. Each group of facial images represents the original image and its corresponding expected output, as well as the watermarked image and its corresponding disrupted output, under the same target label.

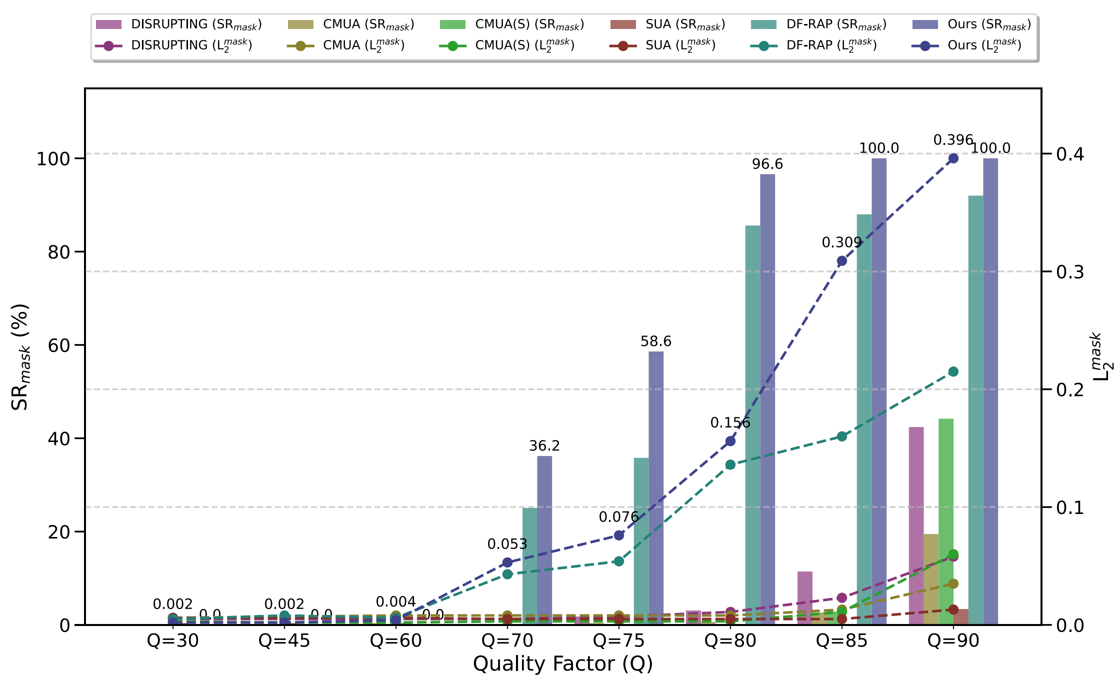

Fig. 4. Comparison of the defense performance of different methods under varying JPEG compression quality factors (Q-values), evaluated against StarGAN with CelebA dataset.

inherent in JPEG compression significantly disrupts the effective distribution of adversarial watermarks, thereby undermining their proactive defense capabilities. Nevertheless, it is worth noting that our method demonstrates notable robustness within the Q-value range of 70 to 90. Considering that this range is commonly adopted on social media platforms (e.g., Q = 75 is the default setting in many applications to balance file size and image quality), we adopt JPEG compression with Q = 75 as the standardized evaluation setting for subsequent experiments.

In this subsection, we compare the proposed method against baseline approaches under both lossless application and lossy compression scenarios. Under lossless conditions, adversarial watermarked images are analyzed directly, without post-processing, to evaluate the defensive performance of all methods. Notably, a JPEG quality factor of 75 — which is the default parameter for JPEG images in most practical image compression applications — is used to simulate real-world lossy compression effects. Specifically, we compare the proposed method with baselines in both scenarios and calculate their quantitative defense performance metrics against three Deepfake models. In addition, we compare the average generation time of adversarial watermarks required by each method to defend against different Deepfake models.

Table 2
The defensive performance and computational cost metrics comparison with baseline methods. The best result is marked in bold, and the second-best result is underlined. "w/o" and "w/" indicate results without and with JPEG Compression (Q = 75).

Models	Methods	L_2^{mask} ↑(w/o)	SR_{mask} ↑(w/o)	L_2^{mask} ↑(w/)	SR_{mask} ↑(w/)	Aver ↑(SR_{mask})	Time(s)
StarGAN	DISRUPTING (Ruiz et al., 2020)	**1.236**	100.0	0.007	1.8	50.9	2.58
	CMUA (Huang et al., 2022)	0.215	100.0	0.008	0.4	50.2	>2.50
	CMUA(S) (Huang et al., 2022)	0.960	100.0	0.003	0.0	50.0	2.50
	SUA (Qiao et al., 2024)	0.371	100.0	0.008	0.2	50.1	**2.41**
	DF-RAP (Qu et al., 2024)	0.258	100.0	0.064	45.4	72.7	5.19
	Ours	0.489	**100.0**	**0.097**	**76.4**	**88.2**	2.42
AttGAN	DISRUPTING (Ruiz et al., 2020)	0.140	78.4	0.033	19.2	48.9	4.46
	CMUA (Huang et al., 2022)	0.135	91.1	0.011	1.7	46.4	>2.21
	CMUA(S) (Huang et al., 2022)	0.088	64.4	0.003	0.1	32.3	**2.21**
	SUA (Qiao et al., 2024)	0.098	74.8	0.012	1.3	38.1	2.22
	DF-RAP (Qu et al., 2024)	0.079	52.2	0.059	37.6	44.9	10.53
	Ours	**0.233**	**99.1**	**0.074**	**52.6**	**75.9**	2.36
HiSD	DISRUPTING (Ruiz et al., 2020)	0.220	**99.6**	0.006	0.0	49.8	1.31
	CMUA (Huang et al., 2022)	0.111	99.4	0.008	0.0	49.7	>0.83
	CMUA(S) (Huang et al., 2022)	**0.227**	99.5	0.007	0.0	49.8	0.83
	SUA (Qiao et al., 2024)	0.175	99.5	0.005	0.0	49.8	**0.82**
	DF-RAP (Qu et al., 2024)	0.069	57.8	0.044	32.0	44.9	1.24
	Ours	0.092	74.3	**0.063**	**58.8**	**66.6**	0.89

Given the variations in training mechanisms across methods, the values reported in Table 2 represent the average time consumed to protect a single facial image. These values are computed by averaging the total generation time over training images, consistent with the parameter setting T in our proposed method. The CelebA dataset serves as the benchmark for testing, and the L_2^{mask}, SR_{mask} and Time results are summarized in Table 2.

First, in the lossless application of the adversarial watermarked image testing scenario (i.e., without JPEG compression), all baseline methods demonstrated significant defensive efficacy against Deepfake models. For StarGAN, experimental results (see Table 2 indicate that all methods achieved a 100% SR_{mask} with L_2^{mask} values consistently exceeding 0.215 (with Method DISRUPTING reaching 1.236). This observation underscores the inherent vulnerability of the StarGAN model to adversarial attacks. Notably, Method DISRUPTING — leveraging a facial identity-specific watermark generation strategy — exhibited significantly higher L_2^{mask} values compared to universal defense frameworks, suggesting that customized adversarial watermarks can enhance targeted defense robustness against StarGAN.

When tested on the more robust AttGAN model, the proposed method achieved superior performance; its L_2^{mask}(0.233) and SR_{mask}(99.1%) outperformed all baselines, particularly when compared to CMUA and SUA (detailed metrics in Table 2. This performance advantage is attributed to a fundamental divergence in methodology design. Specifically, our approach concentrates optimization resources to maximize defense strength against a single target model. In contrast, CMUA and SUA are constrained by multi-model compatibility requirements, forcing a trade-off between defense generality and depth. Such compromises often result in suboptimal performance in universal adversarial watermarking frameworks, as they risk overfitting to specific attack models while sacrificing efficiency. The quantitative comparisons confirm that our method not only delivers excellent defensive performance but also consistently disrupts the output of target Deepfake models with higher stability. In defending against HiSD, our method demonstrates slightly inferior performance compared to existing baselines (spatial-domain adversarial watermark components), implying that HiSD may possess greater inherent resilience to frequency-domain adversarial watermarking. Nonetheless, the L_2^{mask} value achieved by our approach remains nearly twice the empirical threshold (0.05), underscoring its capacity to induce a non-negligible level of adversarial watermark and confirming its partial effectiveness even under more robust model conditions. Due to the inherently higher robustness of AttGAN and HiSD against adversarial attacks and their more complex network architectures compared to StarGAN, the ComGAN mechanism introduced in DF-RAP further disrupts the optimization focus required for effective adversarial perturbation generation in proactive defense. As a result, DF-RAP exhibits comparatively weaker quantitative performance when defending against these two models.

To further validate the proposed method's practical robustness, we applied JPEG compression to adversarial watermarked images, thereby simulating real-world lossy compression during transmission. As illustrated by the visualization results in Fig. 5, all methods except DF-RAP and the proposed approach fail to maintain distinguishable differences between protected output and forged images after undergoing JPEG compression. Notably, the proposed method induces more prominent disruptions in Deepfake-targeted regions such as facial features and hair, which demonstrates its stronger capability to selectively obstruct attribute-specific manipulations introduced by Deepfake models. We then compared the degradation in defensive performance across baseline methods. Experimental results demonstrate that the defensive efficacy of adversarial watermarks generated by baseline approaches deteriorates significantly after JPEG compression. As shown in Table 2, the SR_{mask} metrics for all baselines exhibit severe degradation relative to the uncompressed scenario. For example, Method DISRUPTING, which achieved an L_2^{mask} of 1.236 without compression, experienced a decline to 0.007 in L_2^{mask} and a drop to 1.8% in SR_{mask} after compression; similarly, this significant performance degradation is especially pronounced in the defense against HiSD, where the application of JPEG compression leads to a substantial degradation of defensive performance in other baseline methods, rendering them nearly ineffective. Nevertheless, only DF-RAP manages to retain partial defensive effectiveness, which can be attributed to the ComGAN-guided optimization of more robust adversarial perturbations. Despite this, its overall defense efficacy remains relatively limited. This phenomenon is attributed to the spatial-domain adversarial perturbation generation strategy employed by the baselines, whose high-frequency sensitivity inherently conflicts with the DCT-based quantization mechanism of JPEG compression. Specifically, the quantization process alters the intensity of high-frequency image components, thereby disrupting the effective distribution of adversarial watermarks and substantially undermining their defensive capability.

In contrast, the proposed method maintains stable defensive performance even under JPEG compression, owing to its integrated pre-quantization simulation mechanism during training and its frequency-domain adversarial watermark design. As detailed in Table 2, our method achieves L_2^{mask} values consistently above the 0.05 threshold (specifically, 0.097 for StarGAN, 0.074 for AttGAN and 0.063 for HiSD) and outperforms all baselines in SR_{mask} for all Deepfake models. Furthermore, the data presented in Table 2 (see Time(s)) demonstrate that

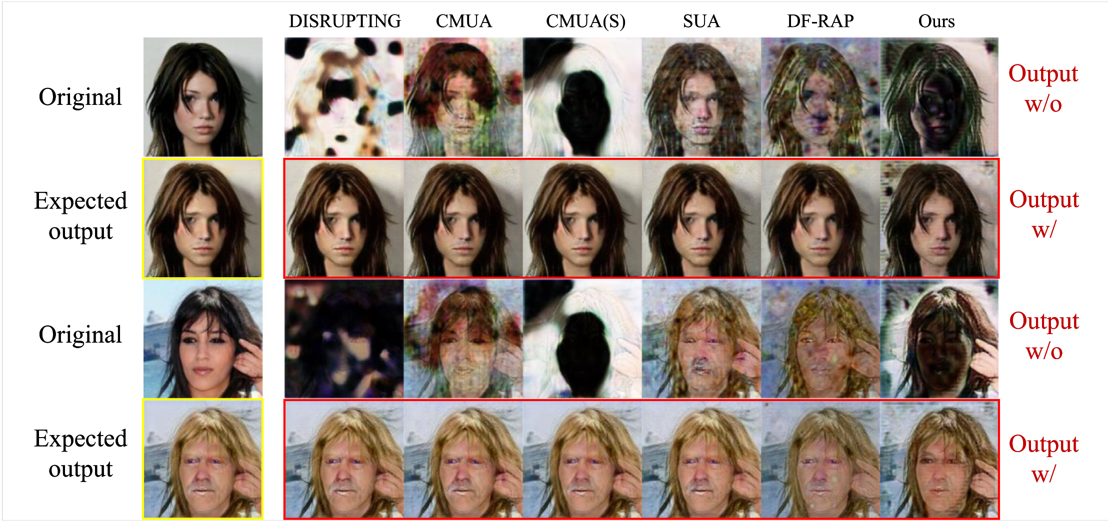

Fig. 5. Visual results of different methods with (w/) and without (w/o) JPEG compression (Q=75). Yellow boxes indicate forged outputs, while red boxes highlight protected image outputs after JPEG processing. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

universal adversarial watermarking approaches, including CMUA(S), SUA, and the proposed method, achieve superior training efficiency. In contrast to single-image protection methods such as DISRUPTING and DF-RAP, these universal methods significantly reduce computational time. It is worth noting that CMUA incurs substantially higher computational overhead — both in terms of processing time and resource consumption — due to its multi-model collaborative optimization strategy, as opposed to configurations designed for single-model protection. In addition, DF-RAP introduces ComGAN into its framework, which significantly increases its computational complexity. These results confirm that the adversarial watermarks generated by our method provide reliable proactive defense regardless of JPEG compression, demonstrating superior advantages in real-world compression scenarios.

4.3.3. Performance in cross-image setting

We validated the defense effectiveness of the proposed method against Deepfake on diverse facial image types through cross-dataset setting, while also verifying its robustness under JPEG compression (quality factor = 75). Specifically, adversarial watermarks were trained and generated on a benchmark dataset and then evaluated for defensive efficacy on facial images from an independent test dataset. Qualitative and quantitative results are presented in Fig. 6 and Table 3, respectively.

Experimental results indicate that under uncompressed defense scenarios, the proposed method exhibits superior performance in cross-dataset testing against three Deepfake generative models. Notably, adversarial watermarks trained on the CelebA benchmark dataset achieve peak performance — attaining an L_2^{mask} of 0.515 and an SR_{mask} of 100.0% — when protecting LFW facial images against StarGAN attacks. In contrast, adversarial watermarks generated from the LFW dataset yield an SR_{mask} of 99.5% (with an L_2^{mask} of 0.302) when defending CelebA facial images against AttGAN-based manipulations. As demonstrated by the comparative data in Tables 1 and 3, the proposed method not only achieves highly effective defense on the original training datasets but also provides reliable cross-dataset protection, confirming its broad applicability to diverse facial image types.

Furthermore, under JPEG compression scenarios, the method retains significant cross-dataset robustness. It achieves L_2^{mask} values exceeding 0.104 (with an SR_{mask} of 81.3%) against StarGAN attacks and L_2^{mask} values exceeding 0.079 (with an SR_{mask} of 65.0%) against the AttGAN model. Notably, in cross-dataset evaluations on HiSD, our method maintains L_2^{mask} values near the 0.05 threshold even when subjected to JPEG compression, demonstrating robust defensive behavior under lossy conditions. These results validate the method's stability in compressed environments, demonstrating that conventional JPEG conversion does not lead to catastrophic failure of defensive efficacy, nor does it necessitate training adversarial watermarks independently for specific facial images. The dual robustness in both cross-dataset and JPEG-resistant scenarios positions the proposed method as a practical solution for large-scale facial image protection tasks.

4.4. Ablation study

In this section, we first analyze the selection and effectiveness of critical hyperparameters, followed by a comparative study on the impact of enhanced fusion strategies on the cross-image defense performance of the proposed adversarial watermarks. As illustrated in Fig. 7, two distinct curves depict the relationship between the hyperparameter q (which controls the JPEG compression strength applied during training) and two key metrics: (1) defensive performance under JPEG compression and (2) the visual quality of adversarial watermarked images, both measured by the average value. Notably, these curves reveal that the choice of q simultaneously governs both the defensive efficacy and the human perceptual quality of the generated adversarial watermarks. Specifically, lower q values (e.g., $q < 30$) improve defensive performance — as evidenced by elevated L_2^{mask} values — but severely degrade visual quality (PSNR < 30 dB). Conversely, higher q values (e.g., $q > 50$) preserve image fidelity (PSNR > 32 dB) at the expense of reduced robustness to JPEG compression. To balance these competing objectives, we empirically set $q = 35$ based on the trade-off trajectory derived from the two curves.

To further validate the effectiveness of the fusion strategy, we perform a comparative analysis of its operational mechanisms. In the absence of the fusion strategy during training, the gradients of single- and multi-batch adversarial watermarks are linearly summed via PGD optimization rather than using the designed fusion strategy. Adversarial watermarks were trained using CelebA as the benchmark dataset and evaluated on LFW via cross-dataset testing. As shown in Table 4, experimental results indicate that this approach leads to persistent cross-watermark gradient conflicts, which manifest as two critical flaws: a significant degradation in visual stealthiness (PSNR = 22.76 dB) and a severe impairment of defense capability (with L_2^{mask} dropping to 0.101 and SR_{mask} falling below 84.0%). Conversely, adversarial watermarks generated using the fusion strategy maintain imperceptibility (PSNR > 30 dB) while achieving targeted protection of facial attributes, effectively preventing Deepfake models from tampering with specific facial

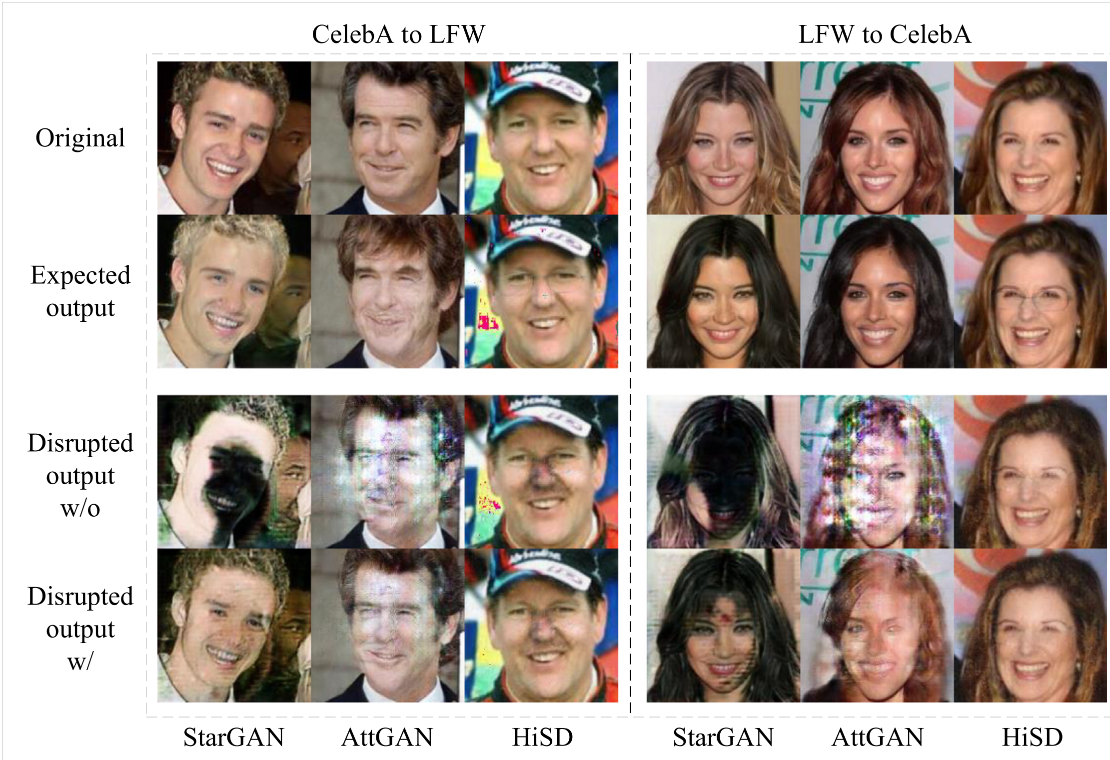

Fig. 6. The image samples from the cross-image experiments, each column shows adversarial watermarks generated from a benchmark dataset (distinct from the test dataset) and applied to the original image. The figure displays both the expected outputs from three Deepfake models and the disrupted outputs after watermark embedding. "w/" and "w/o" denote results with and without JPEG compression (Q = 75), respectively.

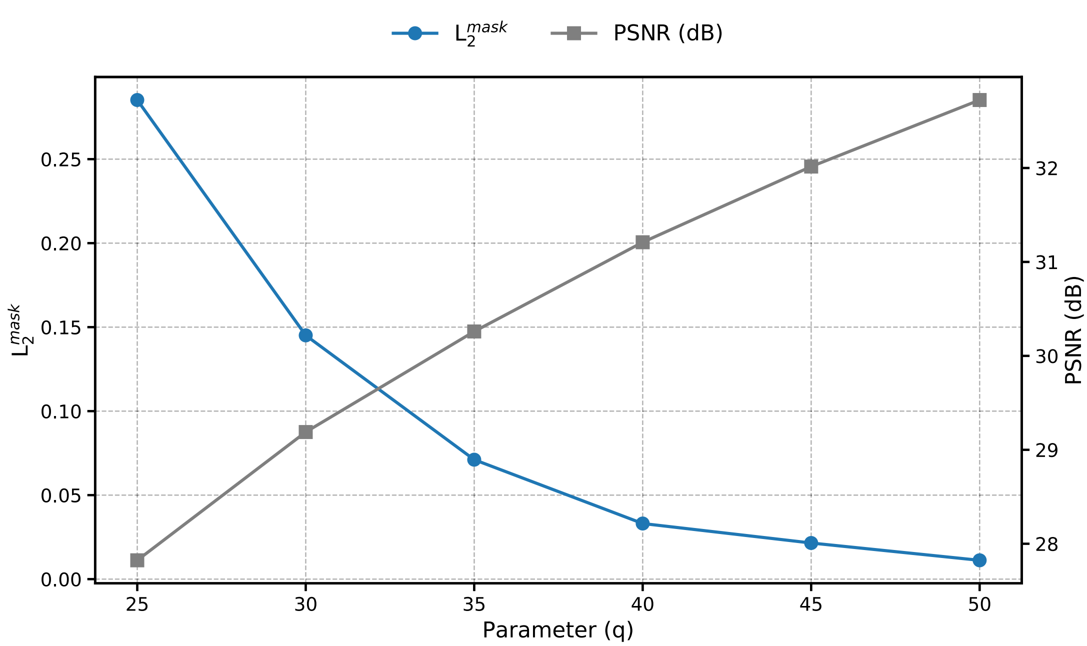

Fig. 7. Disruption performance and Imperceptibility of the adversarial watermark under JPEG Compression (Q = 75) with different values of q, and StarGAN is adopted for evaluation.

Table 3
The defensive performance metricx L_2^{mask} and SR_{mask} in cross-dataset testing. "w/o" and "w/" indicate results without and with JPEG Compression (Q = 75).

Benchmark Dataset	StarGAN				AttGAN				HiSD			
	w/o		w/		w/o		w/		w/o		w/	
CelebA	0.515	100.0	0.104	81.3	0.209	99.0	0.083	70.6	0.060	53.6	0.058	55.2
LFW	0.384	100.0	0.114	86.7	0.302	99.5	0.079	65.0	0.056	50.4	0.044	34.4

Table 4
Ablation results for the fusion module.

Fusion module	×	✓
Imperceptibility (PSNR)↑	22.76 dB	30.10 dB
Defensive (L_2^{mask} /SR_{mask})↑	0.101/84.0%	0.515/100.0%

features. Cross-dataset testing results, as presented in Table 4, further indicate that the proposed method achieves an SR_{mask} of 100.0% across different dataset distributions, confirming that the fusion strategy effectively coordinates cross-sample perturbation patterns to enhance defense generalization.

5. Conclusion

This study proposes an adversarial watermark generation method with enhanced robustness to JPEG compression, enabling proactive defense against Deepfake for large-scale facial images through frequency-domain optimization. The core innovation lies in transforming traditional pixel-domain adversarial watermark training into a frequency-domain optimization problem, where quantization and compression effects are simulated during training via a differentiable JPEG module. In addition, adversarial attack algorithms are integrated to generate frequency-domain adversarial watermarks, while a fusion strategy is designed to coordinate watermark distributions across diverse facial images, thereby improving the method's generalization capability. Experimental results demonstrate that the generated adversarial watermarks maintain significant defensive performance under JPEG compression. Notably, the method eliminates the need for retraining and can be directly applied to different facial images, making it highly practical for real-world scenarios requiring defense against Deepfake models. In future work, we plan to extend our framework to diffusion-based architectures by redesigning the perturbation embedding mechanism to operate over the iterative sampling process used in diffusion models.

CRediT authorship contribution statement

Zhiyu Lin: Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Hanbin Lin:** Writing – original draft, Software, Methodology, Data curation. **Liqiang Lin:** Writing – original draft, Software, Data curation. **Shuwu Chen:** Writing – review & editing, Funding acquisition, Formal analysis. **Xiaolong Liu:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the funds of Project of Industry-university-institute Cooperation in Colleges and Universities in Fujian Province (grant no. 2024h6007, 2024h6030), the Project of Software Technology Innovation Research and Industrialization in Fujian Province (Grant No. 2024.488), and project on digital transformation of higher education (grant no. gjx25z2111) by the employment association of college graduates.

Data availability

Data will be made available on request.

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